

Synthetic CT Generation from MRI for Radiotherapy Applications

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Abstract

This project is centered around synthesizing Computed Tomography (CT) images from Magnetic Resonance Imaging (MRI) data for pelvic radiotherapy treatment planning, aiming to facilitate an "MR-only" workflow to reduce patient radiation exposure and improve treatment efficiency.

Our models were trained using the SynthRAD2023 dataset (180 patients), and more precisely on T1-weighted and T2-weighted MR images of the pelvis. Data preprocessing involved isotropic resampling, intensity scaling of both MR and CT to $[-1, 1]$, and extraction of 2D 256x256 axial slices from centered 3D patches. Our first model is a simple 2D U-Net architecture, based on Ronneberger et al. [1], which was selected and trained using L2 to generate synthetic CT (sCT) images from MR images. Our second model is a Poisson flow generative model, based on a UNet architecture that incorporates residual SwinFIR Transformer blocks [2].

The SwinFIR-PFGM model we trained can generate synthetic CT that achieves satisfying quantitative results with MAE 0.11 ± 0.03 and PSNR 20.68 ± 1.84 , surpassing our baseline 2D UNet quantitative results with MAE 0.31 ± 0.06 and PSNR 13.24 ± 1.00 . Here is the link to our code : <https://github.com/Lloydkamole/Cs230-mri-to-ct>

1 Introduction

Radiotherapy use is gaining importance in cancer treatments over time. It uses high-energy radiation to destroy cancerous cells with millimetric accuracy doses. It does so while reducing radiation to nearby healthy tissues and vital organs. Dose accuracy has to be precise to make more accurate treatment plans. It is only achievable by having a precise image of the patient's anatomy, so that the radiotherapy treatment devices can precisely target tumors in the patient's body, while minimizing damage to organs at risk (OARs).

Modern radiotherapy planning uses multiple imaging devices to capture complementary anatomical information. Computed Tomography (CT) provides electron density values necessary for dose calculation and is the standard

imaging modality for treatment planning. Magnetic Resonance Imaging (MRI) offers superior soft-tissue contrast, making it very useful for tumor detection and identification of adjacent OARs, especially in anatomically complex regions such as the pelvis. In current clinical environments, both CT and MRI are commonly used. CT exposes patients to severe radiation, and using these two modules increases the operation cost and time.

In this project we aim to try to conduct the direct synthesis of CT images from MRI data in the pelvic region in order to overcome these restrictions. An MR-only radiation approach might be made possible by producing high-quality synthetic CT (sCT) from MRI. This would simplify the clinical pipeline, remove the requirement for CT acquisition, and avoid inter-modality registration issues.

However, generating realistic sCT images is challenging because MRI and CT encode different physical properties: MRI emphasizes soft-tissue contrast, whereas CT captures variations in electron density, revealing structures such as bone and air that are not directly inferable from MR intensity values. These differences make the MR-to-CT mapping a challenging supervised learning problem. To tackle this problem, we experimented with two complementary architectures: a simple 2D U-Net [1] and a PFGM++ generative model [3].

The U-Net serves as a strong baseline for paired image-to-image translation, using the aligned MR/CT dataset on a slice-by-slice basis. Each input sample consists of a T1- or T2-weighted MR patch extracted from the 3D volume and converted into 2D axial slices, and the network predicts the corresponding synthetic CT (sCT) slice, which can later be reassembled into a full volume.

At the same time, we investigated PFGM++, a new score-based generative framework that uses Poisson flow on high-dimensional surfaces to describe the underlying data distribution. PFGM++ could offer a more robust generative formulation that might produce high quality sCT predictions by learning the continuous MR/CT mapping in a generative space inspired by physics, compared to the supervised U-Net.

2 Related work

The synthesis of CT images from MR data (MR-to-CT) isn't a new research area. Multiple early attempts to generate synthetic CT (sCT) from MR have been described before the advent of deep learning:

Bulk density assignment methods were one of the initial strategies devised to tackle this problem [4, 5]. This typically involve segmenting the MR image into a few distinct tissue classes, such as air, soft tissue and bone. Once these regions are determined, each class is assigned a predefined Hounsfield Unit (HU) value or electron density value. For example, all voxels identified as bone would be assigned a single, representative HU value for bone. These methods have significant drawbacks: the segmentation process can be long and hard. Furthermore, assigning bulk densities does not account for the natural heterogeneity of tissues, potentially leading to inaccuracies in the generated sCT and subsequent dose calculations.

Another major category of traditional techniques includes atlas methods [6, 5]. These approaches rely on a database ("atlas") of registered MR and CT image pairs from other patients. To generate an sCT for a new patient, their MR image is registered (aligned) to the MR images in the atlas. This registration, often a complex non-rigid deformation, establishes spatial correspondence. The deformation fields obtained from the MR-to-MR registration are then used to warp the corresponding CT images from the atlas to the new patient's MR anatomy, thereby creating the sCT. The primary limitations of atlas methods is their computational cost due to the need for multiple registrations and a lack of robustness when dealing with significant anatomical variations between the new patient and the atlas database, or in the presence of pathologies under-represented in the atlas (e.g., some abnormal tumors, etc.).

The rise of deep learning, particularly Convolutional Neural Networks (CNNs), has significantly advanced the field of MR-to-CT synthesis [7], largely outperforming these traditional methods listed above.

The U-Net architecture [1] has been tried many times for paired image translation in medical imaging. Its distinctive encoder-decoder structure with skip connections enables the effective combination of contextual information with fine spatial details which enhance anatomical accuracy. Many U-Net variants have been proposed, including 2D and 3D U-Nets for synthetic CT generation [8].

More recently, physics-inspired generative models such as diffusion models and Flow Matching models have emerged for medical image reconstruction and synthesis. PFGM++ [9, 10, 3] is a generalization of Poisson flow models that uses electrostatics-driven sampling and a score-based diffusion methods to improve robustness compared to classical diffusion models. We chose to im-

plement this achitecture in combination with the SwinFIR transformer architecture [2] for our task.

3 Methods

3.1 Dataset and Pre-processing

The dataset used for this project is the "SynthRAD2023 Grand Challenge dataset: synthesizing computed tomography for radiotherapy," obtained from Zenodo (<https://zenodo.org/records/7260705>) [14]. The dataset contains 180 patient cases, each with paired Magnetic Resonance (MR) and Computed Tomography (CT) pelvis scans. The MR data includes either T1-weighted or T2-weighted sequences. All images are provided in the NIfTI format (.nii.gz), and are represented as 3D volumetric data. 130 patients have been allocated for training, 32 for validation, and 18 for testing after an initial split of the 180 cases.

The data goes through a simple preprocessing pipeline that was applied to standardize the data. First, isotropic resampling to 1x1x1 mm voxel size was performed using bilinear interpolation. Then, the intensity values of both MR and CT scans are scaled to [-1, 1]. Then, 2D slices of 3D volume, each of size 256x256, are extracted from the center of those images.

3.2 Experimental setup/Training configuration

Our experimental setup involved training our 2D U-Net and Transformer-PFGM++ model on a NVIDIA RTX A5000 GPU. The primary software libraries used were PyTorch for the deep learning framework, MONAI for medical images-specific data handling and transformations, NumPy for numerical operations, Nibabel for NIfTI file loading and visualization, and skimage to compute some metrics.

Several hyperparameters were configured for the U-Net and SwinFIR-PFGM training. Multiple learning rates were tried (e.g., 1e-2, 1e-3, 1e-4, 1e-5); 1e-3 ended up being selected for the U-Net, and 1e-4 with learning rate scheduling for the SwinFIR-PFGM. An Adam optimizer was employed. The 2D UNet was trained for 50 epochs, and the SwinFIR-PFGM model for 66 epochs (the loss decreased up to 1.8e-2).

3.3 2D U-Net Architecture

Our UNet network processes a single-channel input MR slice and outputs a single-channel synthetic CT (sCT) scan. Here is a small description of its different parts:

1. Encoder (Downsampling Path): Initially, a double convolutional block (two sequential 3×3 convolutions,

Batch Normalization, ReLU) processes the input, expanding channels (e.g., to 64). This is followed by 4 downsampling blocks: Each block performs MaxPooling operation (halves spatial dimensions) followed by a double convolutional block (which doubles the feature channels). Then, in the bottleneck part of the network, the final downsampling block output is processed by another double convolutional block.

2. Decoder (Upsampling Path): 4 upsampling blocks are symmetrically reconstructing the image. The upsampling blocks use transposed convolutions to double the spatial dimensions. These feature maps are concatenated with the corresponding feature maps (skip connections) from the encoder path. Then a double convolutional block processes the concatenated features and halves the number of channels. Finally, in the output layer, a final convolution is applied, followed by a sigmoid activation that maps the last feature block to the 1-channel sCT output.

To train, the model, we decided to use a mean squared error (L2) loss to compute our cost function:

$$\mathcal{J}_{L2}(sCT, rCT) = \frac{1}{N} \sum_{i=1}^N (rCT - sCT)^2$$

With sCT being the synthesized CT output, and rCT being the real ground truth CT.

3.4 SwinFIR-PFGM Generative Model

Our best attempt at surpassing the baseline was given by an implementation of the PFGM++ method to our problem. PFGM++ extends the original Poisson Flow Generative Model (PFGM)[9] by introducing D additional augmented dimensions, enabling a continuous interpolation between the PFGM formulation ($D = 1$) and diffusion models ($D \rightarrow \infty$). This adjustable dimensionality allows the model to adapt the generative process to the specific characteristics of the MR-to-CT synthesis task.



Figure 1: PFGM++ unify PFGM and diffusion models

In PFGM++, real data points $x \in \mathbb{R}^n$ are translated into an augmented $(n + D)$ -dimensional space by appending D supplemental coordinates. Noise is then sampled from a hemisphere in this augmented space, and each noisy sample evolves back toward the data surface according to an

ODE inspired by a high-dimensional electric field. The electric field, denoted $E(x)$, is constructed as if there were charges located on the data surface, and its magnitude determines the velocity with which perturbed samples move during generation. Intuitively, the augmented space acts as a potential landscape where clean CT images correspond to equilibrium points and noisy samples are progressively transported toward them.

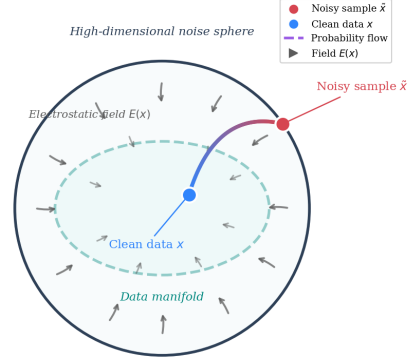


Figure 2: In PFGM++, generation is interpreted as transporting a sample from a noisy high-dimensional sphere toward the data manifold using an electrostatic field. The particle follows a physically motivated probability flow trajectory, rather than being iteratively denoised pixel-by-pixel.

Our SwinFIR-PFGM network processes a dual-channel input (a noisy CT slice and a conditioning MR slice) and outputs a single-channel synthetic CT (sCT) scan. The model is designed as a denoising backbone for a Poisson Flow Generative Model (PFGM++).

1. Encoder (Downsampling Path): Initially, a feature extraction block (GroupNorm, SiLU, 3x3 convolution) processes the input, expanding channels (e.g., to 64). This is followed by 3 downsampling stages. Each stage consists of 2 FIR Residual Blocks and a downsampling layer. The FIR Residual Block is a sophisticated unit that first applies FiLM conditioning (injecting the noise/radius level embedding) to modulate features. It then processes features through two parallel branches: a Spatial Branch (standard residual convolutions) and a Frequency Branch (Spectral Transform using FFT/IFFT for global context), which are fused to capture both local and global structures. Additionally, Swin Transformer blocks (windowed self-attention) are integrated within these blocks [2] at specific resolutions to further enhance long-range dependency modeling. Downsampling is performed using strided 3x3 convolutions (halving spatial dimensions) while doubling the feature channels. In the bottleneck part of the network, the final downsampling output is processed by another FIR Residual Block with Swin Transformer attention.

2. Decoder (Upsampling Path): 3 upsampling stages symmetrically reconstruct the image. Each stage first uses nearest-neighbor interpolation, followed by a convolution

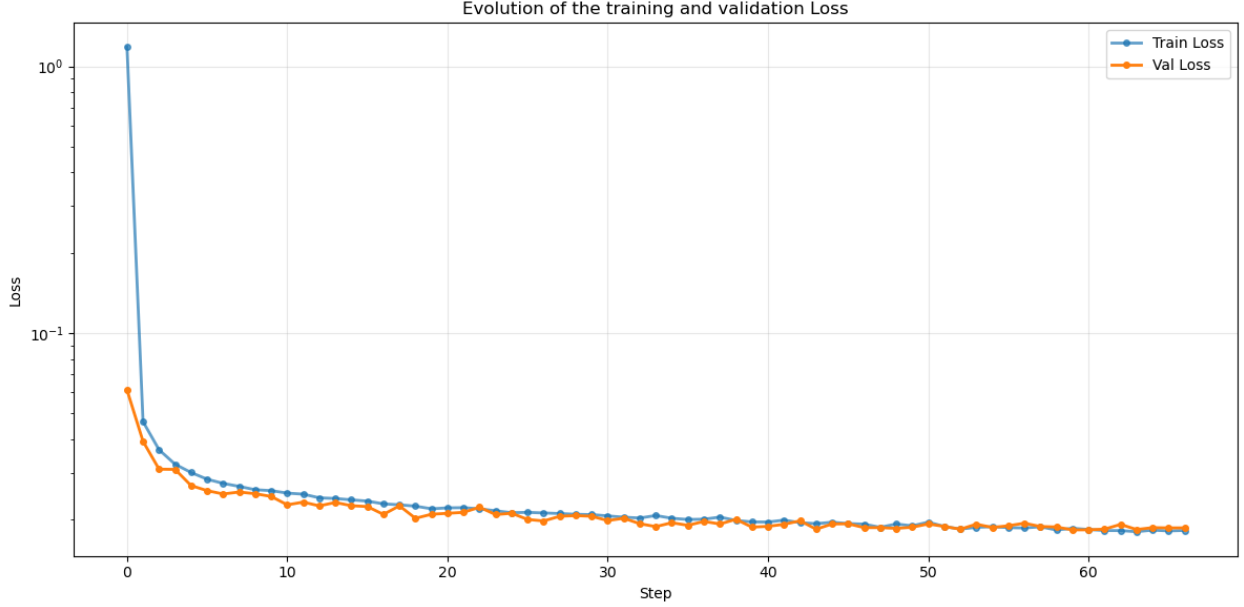


Figure 3: Evolution of the training and validation loss.

to double the spatial dimensions. These upsampled feature maps are concatenated with the corresponding feature maps (skip connections) from the encoder path. Then, a FIR Residual Block (with FiLM conditioning, SFB, and Swin attention) processes the concatenated features and halves the number of channels. Finally, in the output layer, a final convolution is applied to map the last feature block to the 1-channel sCT output.

To train the model, we used the Elucidating Diffusion Models (EDM) framework (<https://github.com/NVlabs/edm>) [11] as an inspiration, and Classifier-Free Guidance, with a weighted mean squared error (L2) loss to compute our cost function, learning to denoise the CT images conditioned on the MRI input.

During training, MR–CT pairs are perturbed with noise at varying radii, and the model learns to predict the direction of the electric field that transports a noisy CT sample back toward its clean counterpart. This corresponds to matching the probability flow induced by the ground-truth CT distribution. During inference, an MR slice conditions the generative process, and the ODE is numerically solved from an initial noisy state to produce a corresponding synthetic CT slice.

4 Experiments/Results/Discussion

Qualitatively, we can see that the synthesized CT images of the two models mimic well the most important features of the real scans, e.g., the location of bones and most soft tissues boundaries like the muscles, are well defined. However, the soft tissues generated by the UNet aren't

as sharp (less detailed) as the ones from the ground truth. The PFGM seems to fare better; the soft tissues generated are sharper and more in line with the ground truth. Nevertheless, both models still create a lot of hallucinations (e.g., soft tissues, air cavities, and metal artefacts that shouldn't be there). The overall boundaries of patients' bodies are respected on both models. Unfortunately, however, the intensity values of anatomical features and boundaries between different types of soft tissue structures aren't always respected, especially for the simple UNet model; the PFGM seems to respect the general intensity values of the ground truth CT scan better.

To qualitatively evaluate the performance of our synthetic CT (sCT) generation, we employed several standard metrics. The Mean Absolute Error (MAE) and the Peak Signal-to-Noise Ratio (PSNR):

$$\text{MAE} = \frac{1}{n} \sum_{j=0}^n |sct_i - rct_i| \quad (1)$$

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{2^2}{\text{MSE}} \right) \quad (2)$$

	2D UNet	PFGM
Mean Absolute Error (MAE)	0,31 ± 0,06	0,11 ± 0,03
Peak Signal-to-Noise Ratio (PSNR)	13,24 ± 1,00	20,68 ± 1,84

Figure 5: Quantitative analysis of synthetic CT

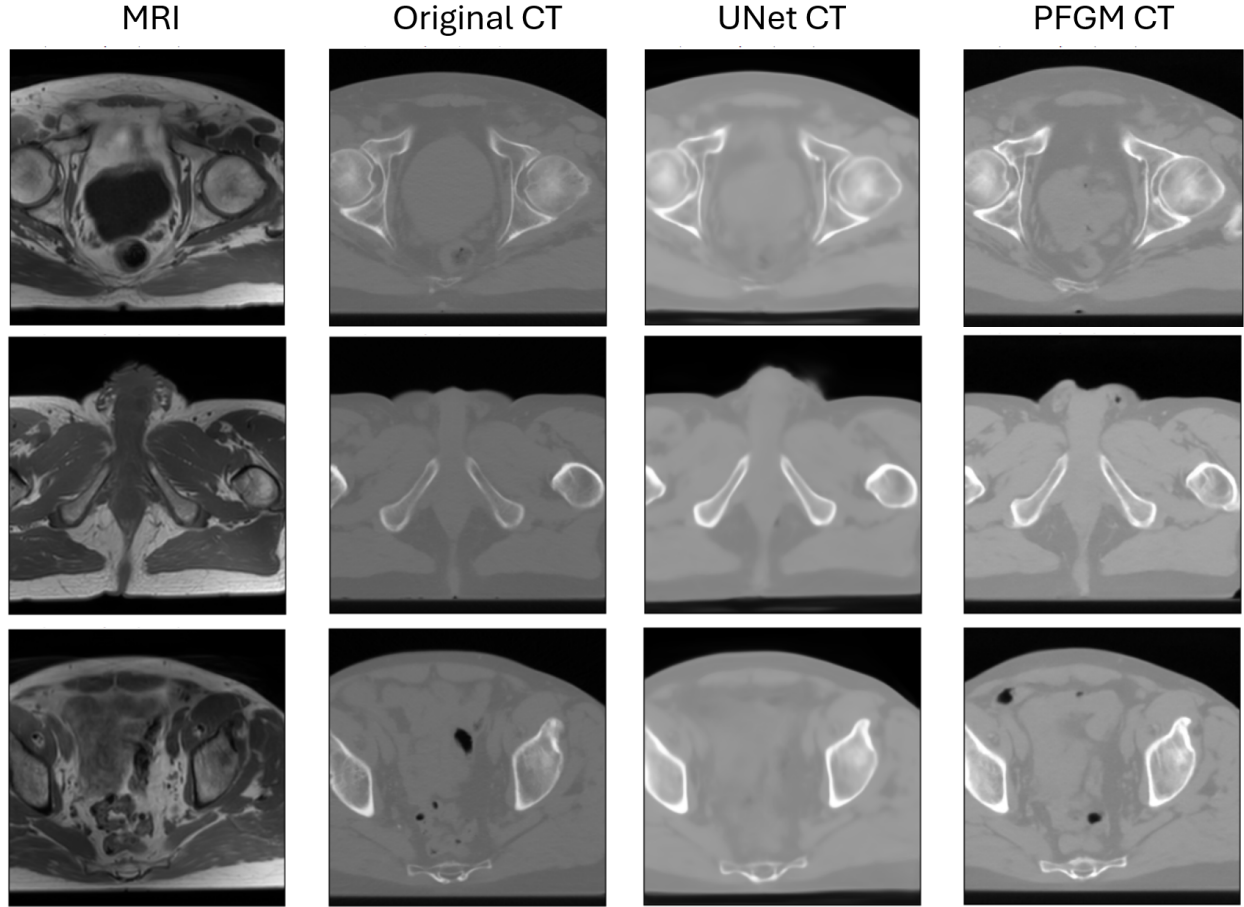


Figure 4: Groundtruth MRI and CT (on the left) and synthetic CTs generated by 2D Unet and PFGM models (on the right) for 3 different cases.

The PFGM model has the lowest MAE and highest PSNR scores; therefore, both qualitative and quantitative results show that the model yields the best synthetic CTs.

Due to the short time allocated by our class, we weren't able to iterate over multiple ideas. Next steps involve tuning the hyperparameters of the Swin-FIR model; perhaps a higher number of FIR residual blocks (e.g., 3 or 4) could help improve generation quality. We could also try to implement 2.5D architecture by feeding a stack of slices, instead of only one 2D slice in the model. It would force the model to learn 3D continuity, reducing "flickering" or inconsistencies between slices in the final volumes.

5 Contributions

Lloyd Kamole Ghoms worked on the data preprocessing pipeline, implementation of both the U-Net and PFGM++ models, experimental design, evaluation, and writing of the report.

Nael Ghoundale contributed to model development and training, assisting with architectural choices, training

configurations, validation of the experimental results and writing of the report..

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